

SUMMERLAND, BC, Canada

WMO: 717680

Lat: 49.5625N

Lon: 119.6487W

Elev: 454

StdP: 95.99

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-10.9	-19.8	0.7	-13.2	-16.7	0.9	-9.9	10.5	4.9	8.9	4.6	3.2	290	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	33.0	17.4	31.2	17.0	29.3	16.5	19.1	29.2	18.1	27.9	17.2	26.9	2.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	11.7	22.2	14.4	10.8	21.4	13.3	10.1	20.7	56.1	29.2	52.9	27.8	50.1	27.0	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.4	5.3	-16.3	36.3	2.8	1.4	-18.3	37.3	-19.9	38.1	-21.5	38.9	-23.5	39.9		
			-17.4	20.5	2.4	1.3	-19.2	21.4	-20.6	22.2	-22.0	23.0	-23.7	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-1.3	0.2	4.7	9.1	14.1	17.5	21.8	21.2	15.9	8.9	2.9	-1.2
	DBStd	8.99	4.55	4.18	3.79	3.20	3.70	3.42	3.42	3.23	3.31	3.23	4.31	4.31
	HDD10.0	1474	350	274	166	55	7	0	0	0	1	59	215	347
	HDD18.3	3480	608	508	422	278	139	54	9	12	88	293	463	605
	CDD10.0	1307	0	0	2	27	134	226	367	348	177	24	1	0
	CDD18.3	272	0	0	0	0	8	31	118	101	14	0	0	0
	CDH23.3	2973	0	0	0	3	114	353	1333	1033	137	0	0	0
	CDH26.7	1104	0	0	0	0	24	108	559	382	31	0	0	0
Wind	WSAvg	2.4	2.2	2.3	2.5	2.4	2.3	2.3	2.4	2.4	2.5	2.6	2.3	
Precipitation	PrecAvg	320	30	20	19	25	34	34	27	30	23	17	29	34
	PrecMax	493	66	51	41	61	93	112	72	82	54	63	66	71
	PrecMin	213	6	4	3	5	8	7	0	2	3	4	9	11
	PrecStd	66	14	13	10	17	19	22	17	18	15	13	14	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	10.5	16.6	23.0	29.5	32.6	35.9	34.6	29.7	20.4	14.0	9.4
		MCWB	4.8	6.2	8.7	12.7	15.7	17.2	18.7	17.7	16.6	13.0	9.9	6.4
	2%	DB	6.4	8.4	14.3	19.9	26.5	29.7	33.6	32.3	26.9	17.8	11.6	7.0
		MCWB	4.0	4.8	7.6	10.7	14.3	16.0	17.6	17.2	15.6	11.3	8.1	4.3
	5%	DB	5.0	6.8	12.4	17.7	23.9	27.2	31.9	30.6	24.6	15.9	9.8	5.4
		MCWB	3.1	3.9	6.7	9.4	13.3	15.3	17.3	17.1	14.8	10.2	7.0	3.3
	10%	DB	3.7	5.4	10.6	15.6	21.6	25.0	30.0	28.8	22.5	14.4	8.5	3.9
		MCWB	2.3	3.0	5.6	8.3	12.1	14.5	16.8	16.5	13.8	9.3	5.9	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	7.2	9.9	13.1	16.6	18.7	20.7	20.4	18.2	14.0	10.3	6.9
		MCDB	7.3	9.6	14.6	21.3	26.0	28.5	31.9	31.0	26.2	18.6	12.9	8.9
	2%	WB	4.2	5.5	8.4	11.4	15.2	17.2	19.3	19.1	16.8	12.5	8.6	4.9
		MCDB	5.9	7.4	13.1	18.3	23.7	26.7	29.8	28.8	24.5	16.4	10.7	6.6
	5%	WB	3.3	4.4	7.2	10.1	14.2	16.1	18.4	18.0	15.6	11.2	7.4	3.6
		MCDB	4.7	6.3	11.3	16.2	22.3	24.9	28.6	27.5	23.1	15.0	9.4	5.0
	10%	WB	2.4	3.3	6.2	8.9	13.0	15.2	17.4	17.1	14.5	9.9	6.1	2.4
		MCDB	3.5	5.0	10.1	14.5	20.2	23.3	27.6	26.7	21.4	13.6	8.0	3.7
Mean Daily Temperature Range	5% DB	MDBR	4.8	6.8	9.0	11.1	12.1	12.0	13.5	13.2	11.5	8.8	5.8	4.7
		MCDBR	5.9	7.7	11.2	14.1	15.6	15.4	16.4	15.7	14.2	10.5	7.2	5.9
		MCWBR	4.3	5.1	6.5	7.3	7.3	6.9	6.9	7.0	7.3	6.1	5.0	4.3
	5% WB	MCDBR	5.5	7.1	10.2	12.6	13.6	13.6	14.5	14.2	13.0	9.4	6.6	5.5
		MCWBR	4.1	4.9	6.3	7.0	6.8	6.4	6.8	6.9	7.0	5.9	4.7	4.2
Clear-Sky Solar Irradiance	taub	0.291	0.290	0.318	0.343	0.374	0.366	0.356	0.366	0.340	0.330	0.305	0.280	
	taud	2.330	2.388	2.408	2.403	2.350	2.400	2.454	2.429	2.486	2.490	2.458	2.398	
	Ebn at Noon	756	863	895	905	884	892	897	871	862	805	742	724	
	Edh at Noon	68	81	96	107	118	113	106	104	88	74	60	56	
All-Sky Solar Radiation	RadAvg	1.01	1.95	3.10	4.57	5.43	5.71	6.59	5.61	3.97	2.27	1.06	0.77	
	RadStd	0.12	0.20	0.30	0.35	0.43	0.51	0.53	0.41	0.35	0.28	0.13	0.09	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.58	-0.79	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page